

CS464 — Introduction to Machine Learning

Final Report

Comparative Analysis of Deep Learning Models and Embedding-Based Machine Learning Model for Hand Gesture Recognition

1. Introduction

This final report documents the work completed by Team 3 for the CS464 course project, which focuses on a comparative analysis of deep learning models and an embedding-based classical machine learning approach for hand gesture recognition. The project focuses on two pretrained convolutional neural networks, ResNet-50 and EfficientNet-B0, under two training regimes: frozen backbone training and full fine-tuning. Additionally, a Support Vector Machine (SVM) classifier trained on deep feature embeddings is evaluated against end-to-end deep learning baselines. The primary research questions guiding this project are:

- How do ResNet-50 and EfficientNet-B0 compare under clean and corrupted conditions?
- Does fine-tuning improve performance over frozen backbone training?
- How robust are these models to Gaussian blur degradation?
- How does an embedding-based classical machine learning model compare to end-to-end deep learning?

All planned milestones for the final report have been successfully completed, and additional experiments including blur robustness testing and SVM baseline evaluation have also been completed.

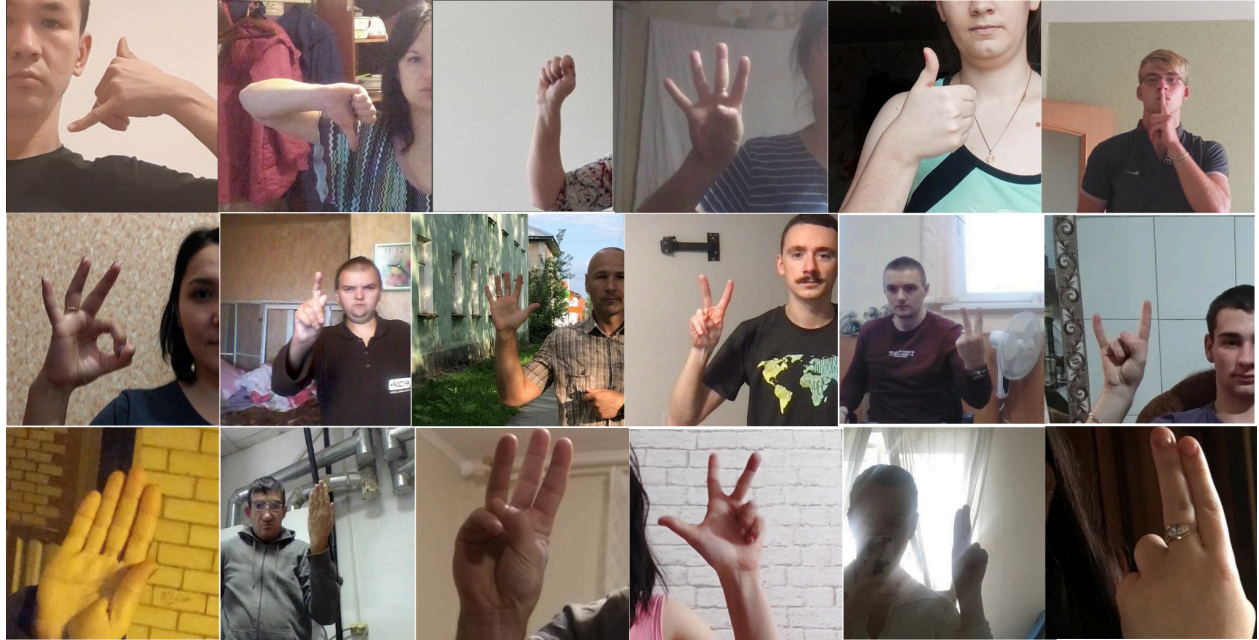
2. Dataset Analysis

2.1 Dataset Overview

The HaGRID Classification 512p dataset was obtained from Kaggle. It consists of 512×512 RGB images of static hand gestures with 18 gesture classes. The dataset was collected under diverse real-world conditions, including varying lighting environments, backgrounds, and hand orientations, making it a realistic dataset for gesture classification.

The 18 gesture classes present in the dataset are:

call, dislike, fist, four, like, mute, ok, one, palm, peace, peace_inverted, rock, stop, stop_inverted, three, three2, two_up, two_up_inverted



2.2 Subsampling Strategy

The full dataset contains approximately 500,000 images. To maintain computational feasibility while preserving class balance, a simple subsampling strategy was applied. Exactly 5,555 images were randomly selected from each of the 18 classes using a fixed random seed (seed=42), resulting in a final working dataset of 99,990 images. The use of a fixed seed ensures full reproducibility of the subsampling process.

Property	Value
Total classes	18
Images per class (original)	~27,000
Images per class (sampled)	5,555
Total sampled images	99,990
Image resolution (original)	512 × 512 pixels
Random seed	42

2.3 Train / Validation / Test Split

Since the HaGRID dataset does not provide predefined train/validation/test splits, the subsampled dataset was divided using a stratified 70/15/15 split. The sklearn train_test_split function was used with random_state=42 to ensure reproducibility and class balance across all three sets.

Split	Percentage	Images per Class	Total Images
Train	70%	3,888	69,984
Validation	15%	833	14,994
Test	15%	834	15,012

3. Dataset Preprocessing

All images were resized to 224×224 pixels to match the input requirements of the pretrained ResNet-50 and EfficientNet-B0 models. Separate transformation pipelines were constructed for the training set and the validation/test sets.

The training set transformation pipeline includes:

- Resize to 224×224 pixels
- Random rotation up to ± 15 degrees
- Color jitter: brightness, contrast, and saturation each varied by ± 0.3
- Conversion to tensor
- Normalization using ImageNet statistics

The validation and test set pipeline does not include augmentation in order to ensure a cleaner evaluation.

4. Coding Environment

All experiments were conducted on a local machine running Windows 11 with the following hardware and software configuration:

Component	Details
GPU	NVIDIA GeForce RTX 3070 Laptop GPU (8 GB VRAM)
CUDA Version	12.1
Driver Version	591.74
CPU	AMD Ryzen 7 5800H 3.2GHz
RAM	32 GB
OS	Windows 11
Python	3.11.9

PyTorch	2.5.1+cu121
torchvision	0.20.1+cu121
timm	1.0.26
scikit-learn	1.8.0
IDE	VS Code with Jupyter extension

The project was developed using a Python 3.11 virtual environment to isolate dependencies. All model checkpoints were saved locally to disk after every training epoch to prevent data loss.

5. Details of the Trained Models

5.1 ResNet-50

ResNet-50 is a pretrained convolutional neural network that has been widely used for image classification tasks. It is known for its strong performance and ability to learn useful visual features from large datasets. In this project, the pretrained ResNet-50 model was adapted for hand gesture classification by replacing its final classification layer with a new layer that outputs predictions for 18 gesture classes. ResNet-50 was used as the main baseline model to evaluate the effectiveness of transfer learning on this dataset.

5.2 EfficientNet-B0

EfficientNet-B0 is another pretrained convolutional neural network designed to achieve high accuracy with fewer parameters compared to traditional models. Similar to ResNet-50, the final classification layer was modified to produce predictions for the 18 gesture classes. EfficientNet-B0 was included in the project in order to compare its performance with ResNet-50 and analyze how different architectures perform on the same hand gesture recognition task.

5.3 Training Configurations

Two training regimes were applied to each architecture:

- **Frozen Backbone:** All pretrained layers were frozen. Only the classification head (fc layer for ResNet-50, classifier layer for EfficientNet-B0) was trained. Optimizer: Adam with learning rate = $1e-3$.

- Fine-tuned: The last residual block (layer4 for ResNet-50, blocks[-1] for EfficientNet-B0) and the classification head were unfrozen. A differential learning rate was applied: learning rate = $1e-4$ for the backbone layers and learning rate = $1e-3$ for the classification head.

Model	Training Mode	Trainable Params	Optimizer	LR (backbone)	LR (head)	Epochs
ResNet-50	Frozen	~36K	Adam	—	$1e-3$	10
ResNet-50	Fine-tuned	~12M	Adam	$1e-4$	$1e-3$	10
EfficientNet-B0	Frozen	~18K	Adam	—	$1e-3$	10
EfficientNet-B0	Fine-tuned	~3M	Adam	$1e-4$	$1e-3$	10

6. Experimental Setup

6.1 Baseline Evaluation (Clean Test Set)

After training, all four models were evaluated on the held-out test set (15,012 images). The following metrics were computed for each model:

- Accuracy: proportion of correctly classified samples
- Macro F1-score: unweighted mean of per-class F1 scores, appropriate for balanced datasets
- Confusion matrix: per-class prediction breakdown visualized as a heatmap
- Per-class precision, recall, and F1 (classification report)

6.2 Robustness Analysis (Blur Corruption)

To evaluate model robustness under visual degradation, Gaussian blur was applied to the test set at three severity levels using kernel sizes of 3, 7, and 11. The blurred test sets were generated. All four models were evaluated on each blur level, and performance degradation relative to clean test accuracy was evaluated.

6.3 SVM Baseline

A classical machine learning baseline was constructed using the best-performing deep learning model (ResNet-50 Fine-tuned) as a fixed feature extractor. Features were extracted from the model for all training and test images. Principal Component Analysis (PCA) was then used to reduce the feature dimension from 2048 to 200 components while preserving most of the useful information. Finally, an SVM classifier with an RBF kernel was trained on the reduced features and evaluated on the test set.

7. Results

7.1 Training Curves Summary

The table below summarizes the best validation accuracy achieved during training for each model. Fine-tuned models show higher performance from the very first epoch compared to frozen variants, confirming the effectiveness of partial unfreezing for transfer learning.

Model	Best Val Acc	Best Epoch	Final Train Acc
ResNet-50 Frozen	61.27%	8	58.66%
ResNet-50 Fine-tuned	97.33%	10	98.40%
EfficientNet-B0 Frozen	56.91%	9	60.07%
EfficientNet-B0 Fine-tuned	91.92%	10	92.92%

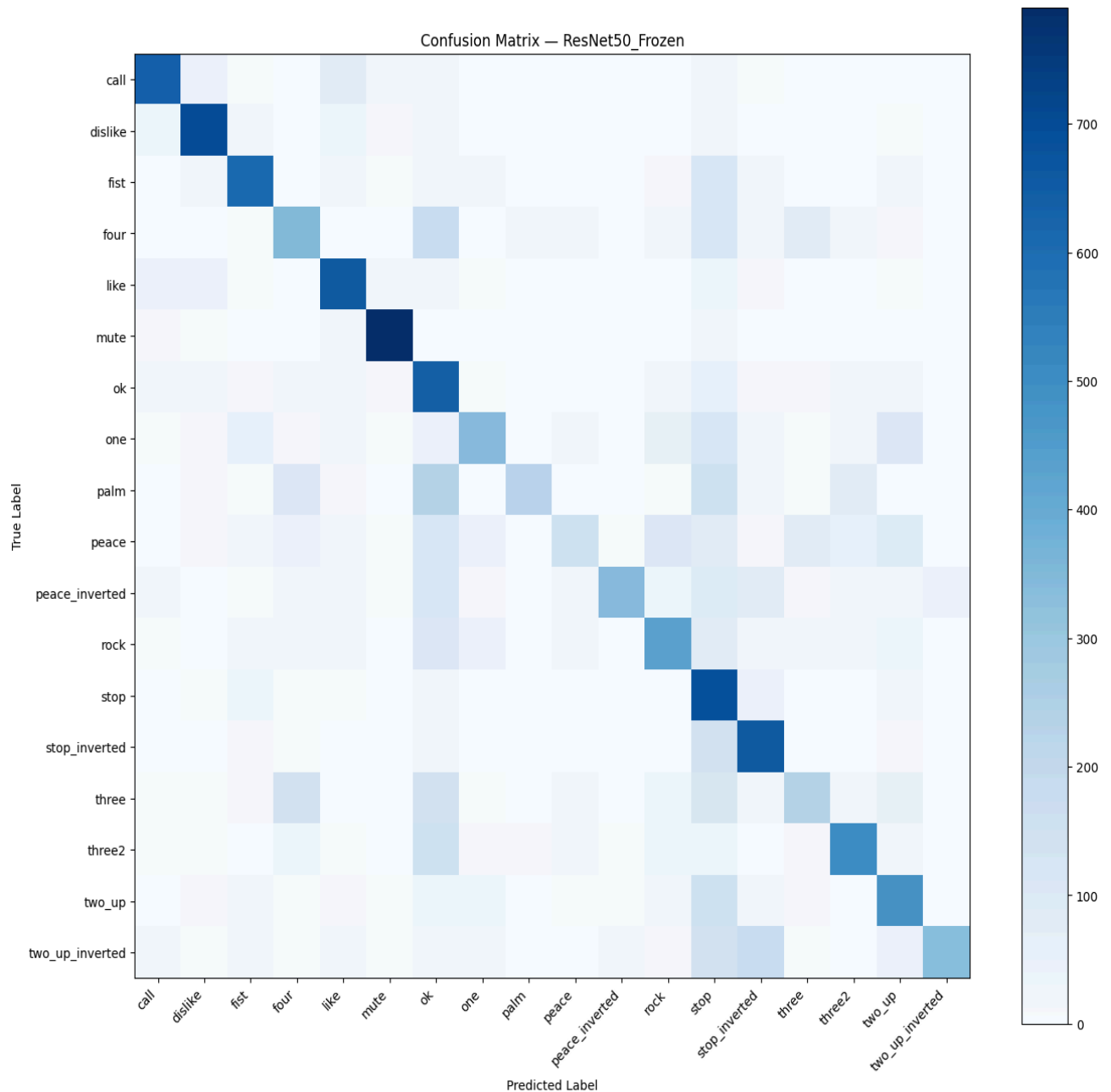
7.2 Clean Test Set Performance

All four models plus the SVM baseline were evaluated on the clean test set of 15,012 images. ResNet-50 Fine-tuned and the SVM achieved the highest performance, with the SVM narrowly outperforming even the end-to-end deep learning model on this metric.

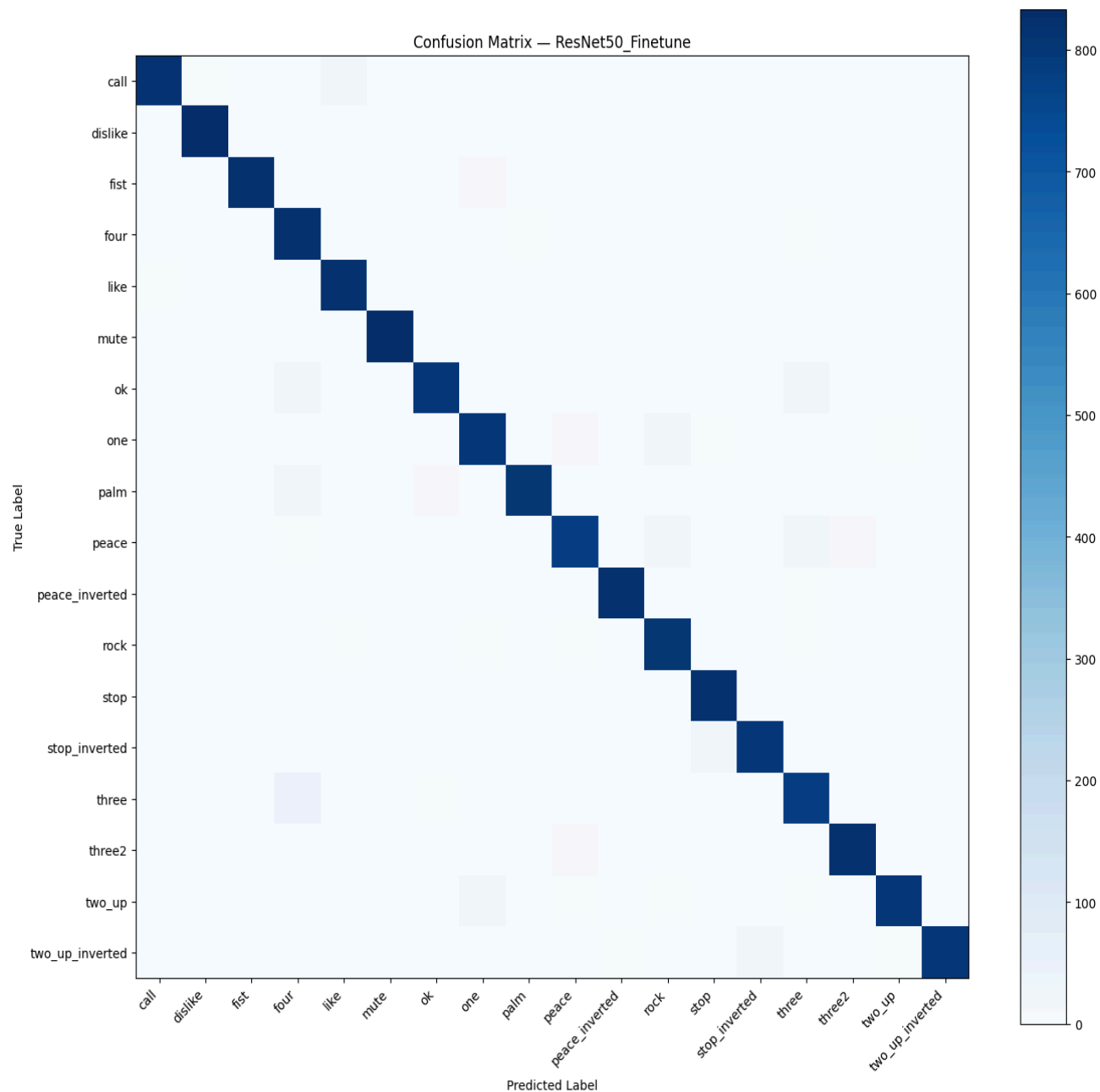
Model	Test Accuracy	Macro F1-Score
ResNet-50 Frozen	58.51%	0.5791

EfficientNet-B0 Frozen	56.52%	0.5618
EfficientNet-B0 Fine-tuned	92.23%	0.9223
ResNet-50 Fine-tuned	97.16%	0.9717
SVM (ResNet-50 Embeddings)	97.67%	0.9767

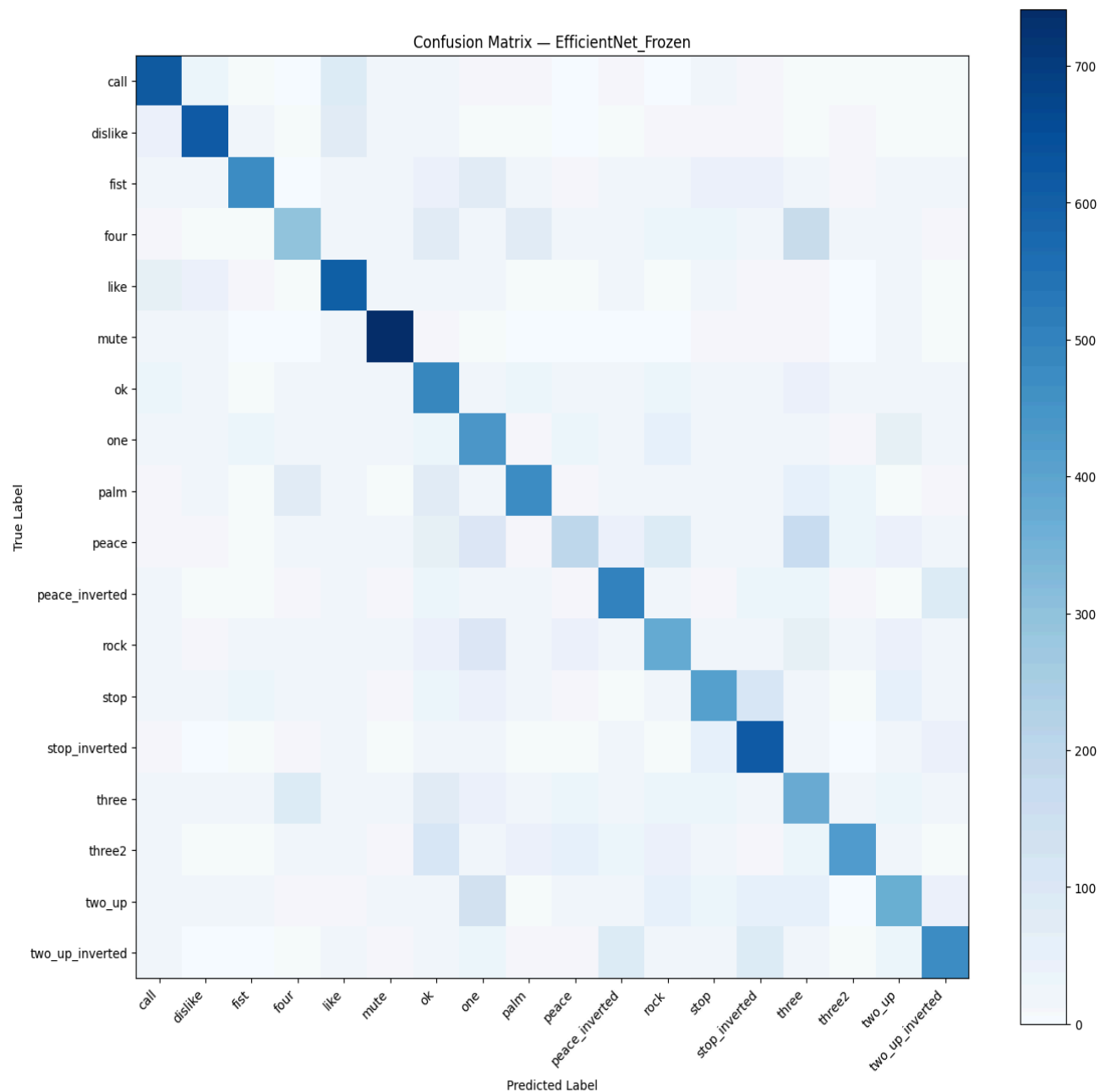
The confusion matrices for each model on the clean test set are provided below in order to visualize class-wise prediction performance:



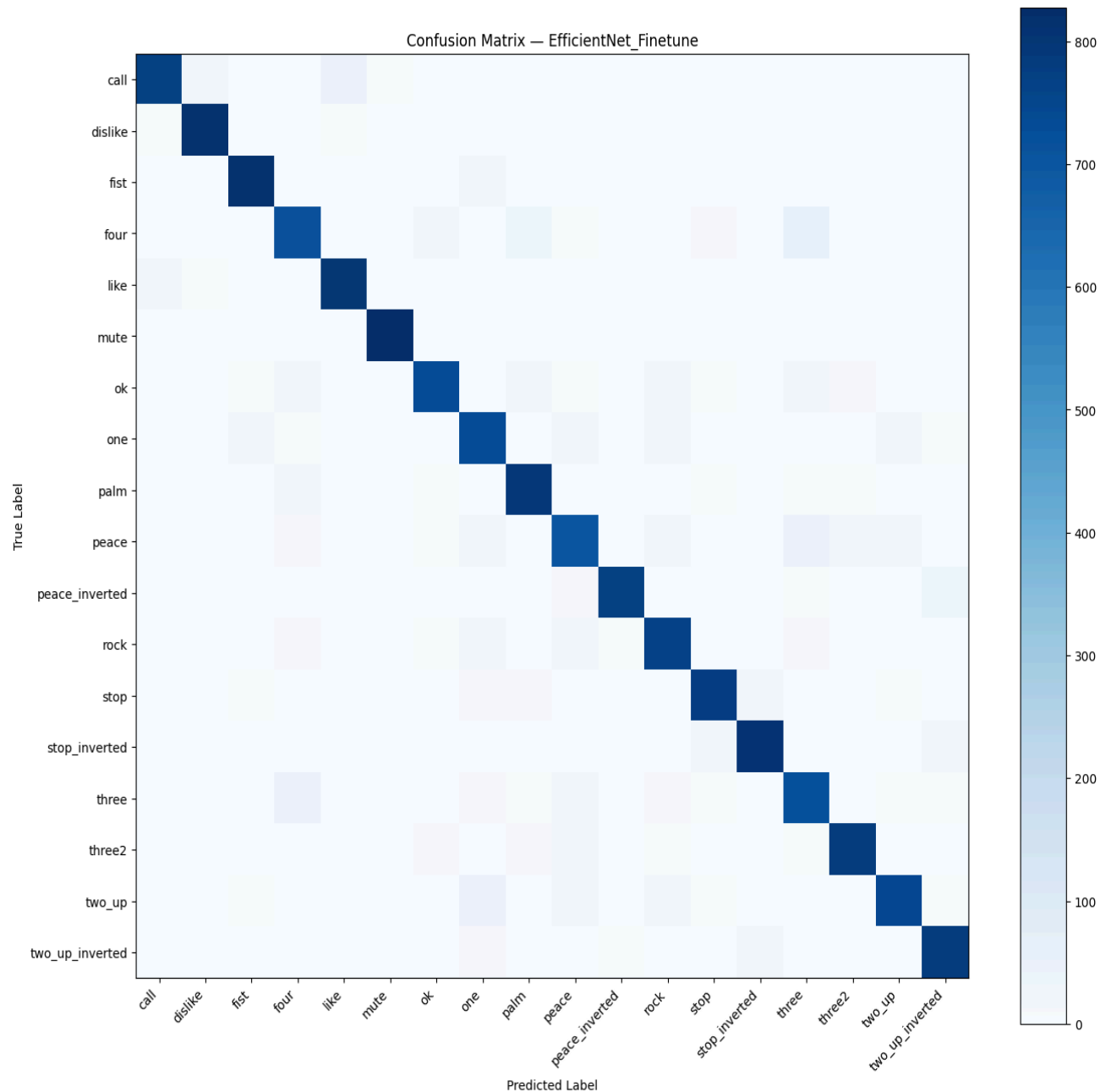
The model is able to correctly classify several gestures, but there is still visible confusion between some classes. Since the network is used without additional training on our dataset, the extracted features are not always specific enough to clearly distinguish similar hand shapes.



The predictions are strongly concentrated along the diagonal, showing that the model distinguishes most gesture classes very clearly. Fine-tuning helps the network adapt its features to the dataset, leading to much more consistent and accurate predictions.



Compared to the fine-tuned models, the predictions are more spread out and some classes are mixed with each other. This suggests that the pretrained features alone are not fully sufficient to separate visually similar gestures in our dataset.



After fine-tuning, the predictions are mostly concentrated along the diagonal, showing that the model can distinguish most gestures quite well. There are still small confusions between a few similar classes, but overall the results indicate a clear improvement compared to the frozen version.

7.3 Frozen vs Fine-tuning Comparison

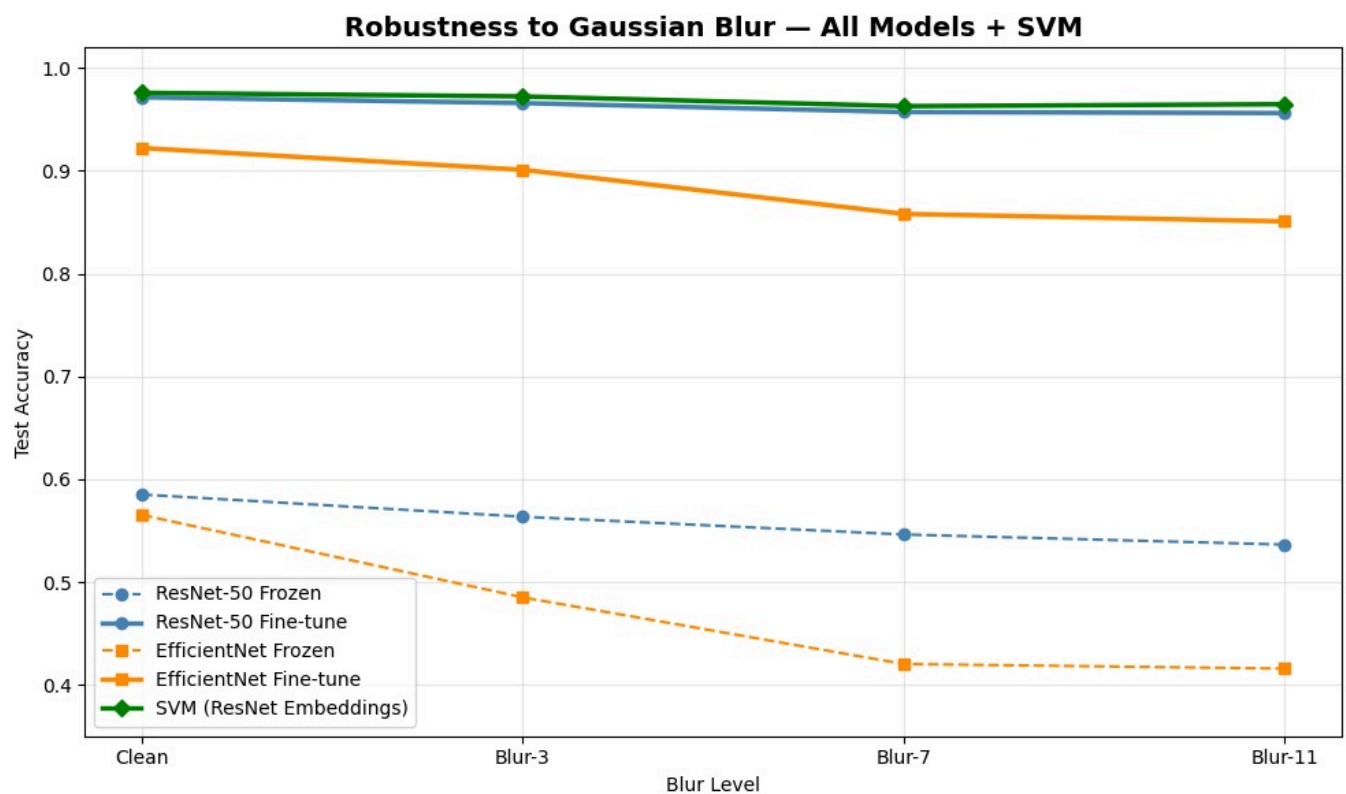
Fine-tuning consistently outperformed frozen backbone training across both architectures. For ResNet-50, fine-tuning improved test accuracy from 58.51% to 97.16%. For EfficientNet-B0, the improvement was from 56.52% to 92.23%. These results suggest that unfreezing the final convolutional block helps the model adapt better to gesture-specific visual patterns than frozen ImageNet features alone. ResNet-50 outperformed EfficientNet-B0 under both training regimes.

Architecture	Frozen Acc	Fine-tuned Acc	Improvement
ResNet-50	58.51%	97.16%	+38.65%
EfficientNet-B0	56.52%	92.23%	+35.71%

7.4 Robustness to Gaussian Blur

Model	Clean	Blur-3	Blur-7	Blur-11	Drop (Clean→11)
ResNet-50 Frozen	58.51%	56.35%	54.62%	53.64%	-4.87%
ResNet-50 Fine-tuned	97.16%	96.60%	95.73%	95.63%	-1.53%
EfficientNet Frozen	56.52%	48.50%	42.02%	41.57%	-14.95%
EfficientNet Fine-tuned	92.23%	90.11%	85.82%	85.09%	-7.14%
SVM (ResNet Embeddings)	97.60%	97.25%	96.30%	96.49%	-1.11%

All models were evaluated under three levels of Gaussian blur corruption. Fine-tuned models showed better robustness than their frozen counterparts, and the SVM baseline had the strongest overall robustness among all evaluated approaches. ResNet-50 Fine-tuned dropped only 1.53 percentage points at the highest blur level (kernel size 11), while the SVM trained on its embeddings dropped just 1.11 percentage points, which is the smallest decrease across all models. This shows that PCA-reduced deep embeddings combined with an RBF-SVM classifier are not only more accurate on clean data but also more resilient to Gaussian blur corruption than any other deep learning model tested. EfficientNet-B0 Fine-tuned showed more sensitivity, dropping 7.14 percentage points at the same blur level. Frozen models, having weaker baseline accuracy, exhibited the largest relative degradation: EfficientNet-B0 Frozen dropped 14.95 percentage points - nearly three times the drop of its fine-tuned counterpart - indicating that frozen depthwise separable convolution features are particularly vulnerable to low-level image corruption when not adapted to the target domain. ResNet-50 Frozen, while also weaker in clean accuracy, showed more moderate degradation at 4.87 percentage points, further confirming the architectural robustness advantage of ResNet over EfficientNet under frozen training conditions.

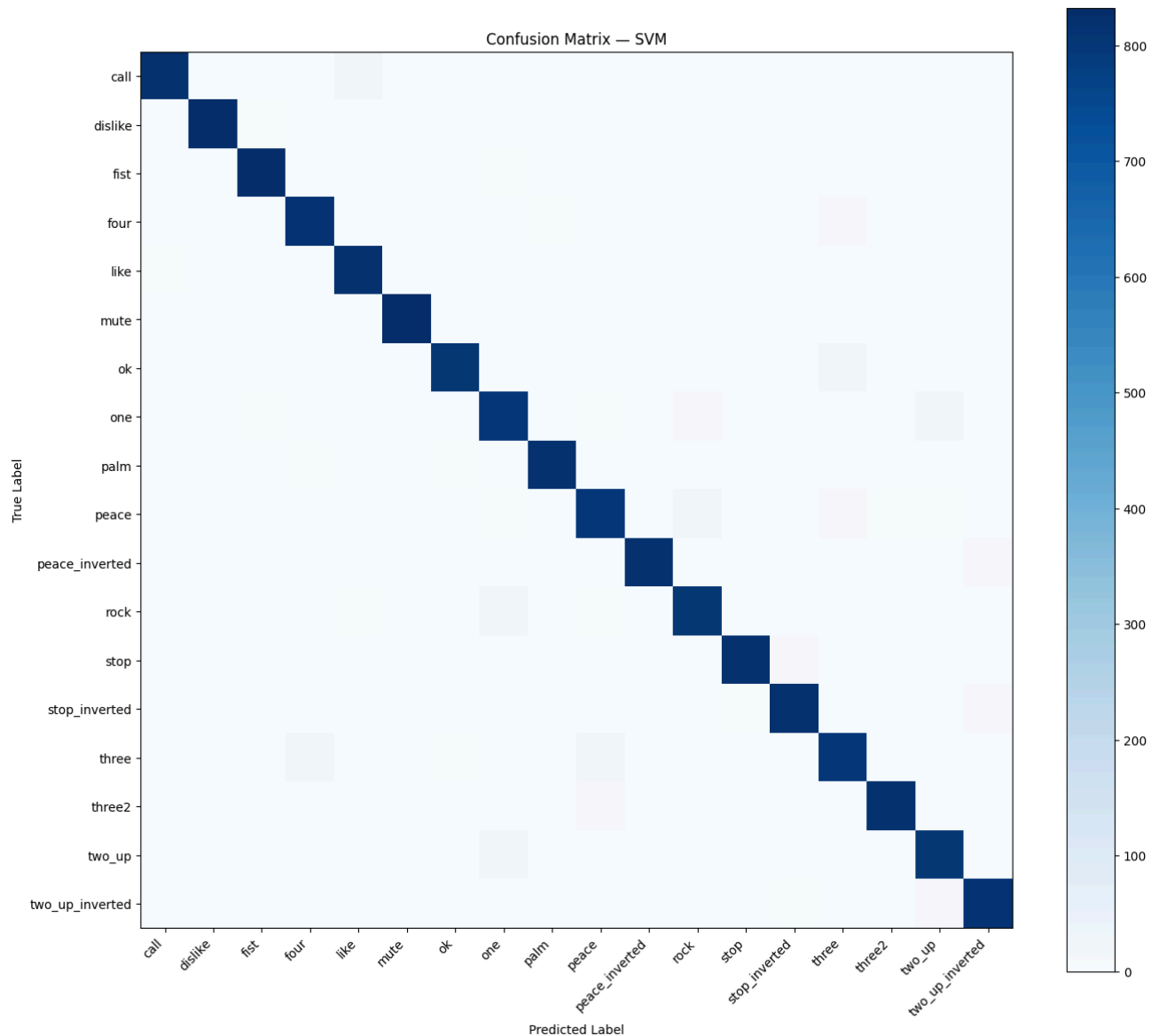


7.5 SVM Baseline Results

The SVM baseline, trained on features extracted from the fine-tuned ResNet-50 model, achieved a test accuracy of 97.67% and a Macro F1-score of 0.9767. This was slightly higher than the performance of the end-to-end ResNet-50 Fine-tuned model on the clean test set.

These results indicate that features extracted from the fine-tuned ResNet-50 model are effective for gesture classification. The SVM classifier was able to achieve slightly higher accuracy than the end-to-end deep learning model on the clean test set. This shows that embedding-based approaches can perform competitively while requiring relatively simple training after feature extraction.

Model	Test Accuracy	Macro F1-Score	Notes
ResNet-50 Frozen	58.51%	0.5791	Frozen backbone
EfficientNet-B0 Frozen	56.52%	0.5618	Frozen backbone
EfficientNet-B0 Fine-tuned	92.23%	0.9223	Last block unfrozen
ResNet-50 Fine-tuned	97.16%	0.9717	Last block unfrozen
SVM (ResNet embeddings)	97.67%	0.9767	Best overall



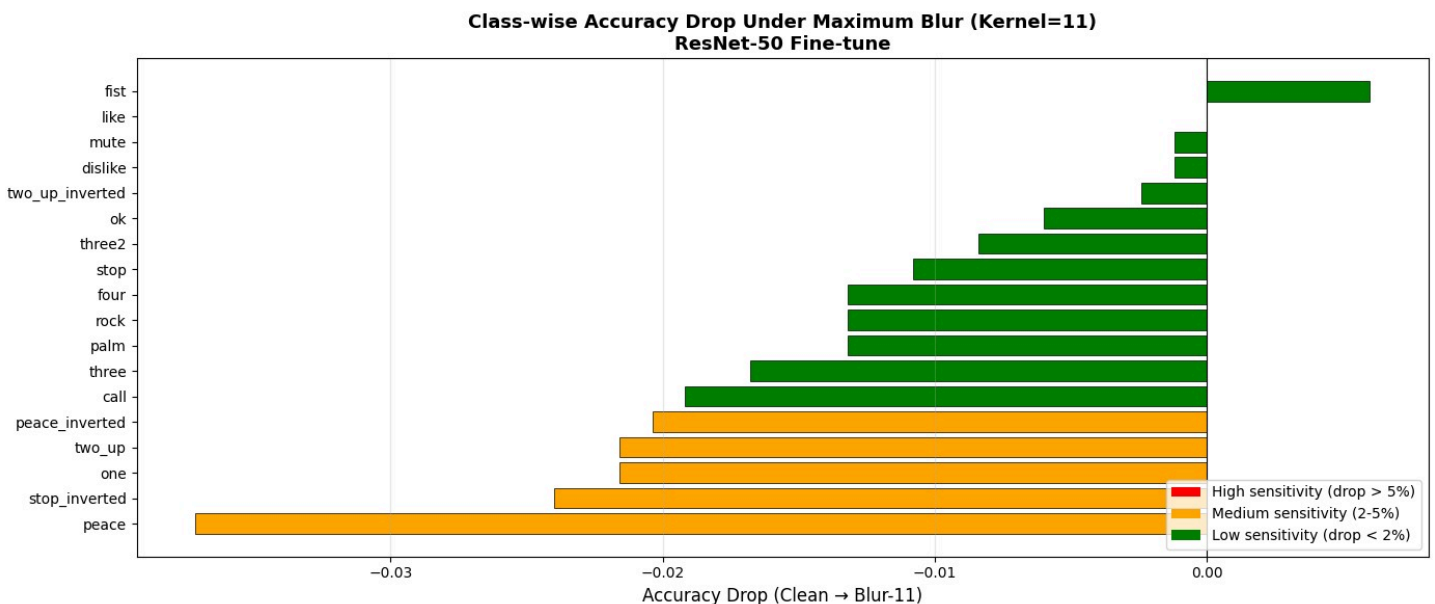
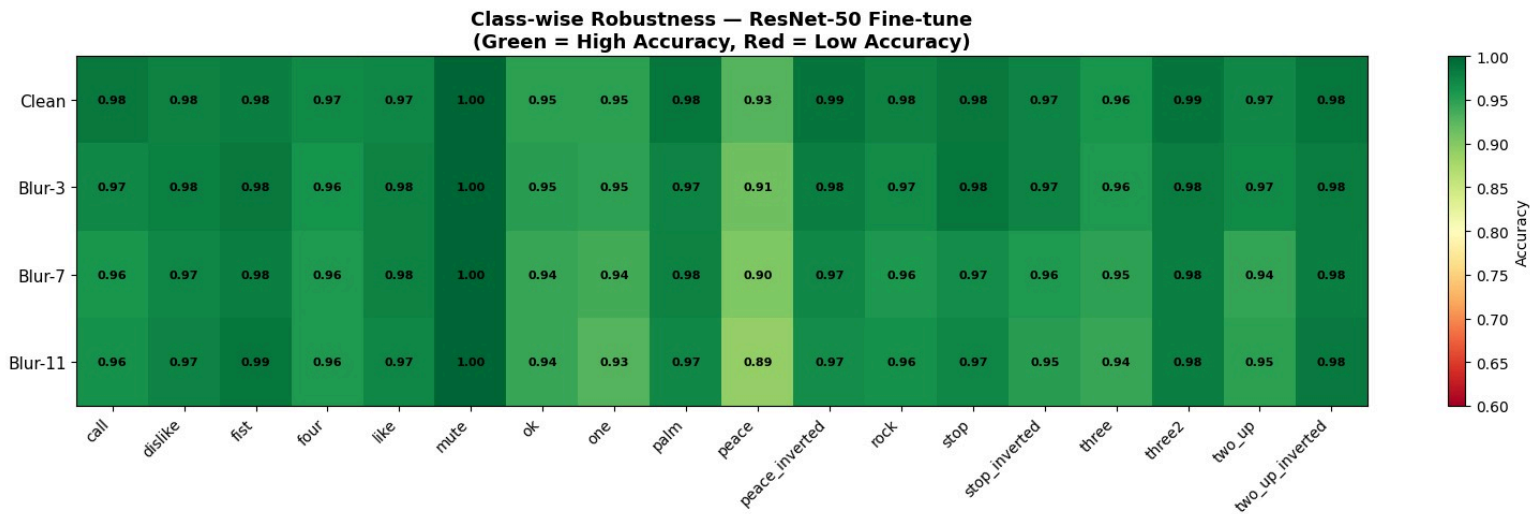
7.6 Per-Class Analysis

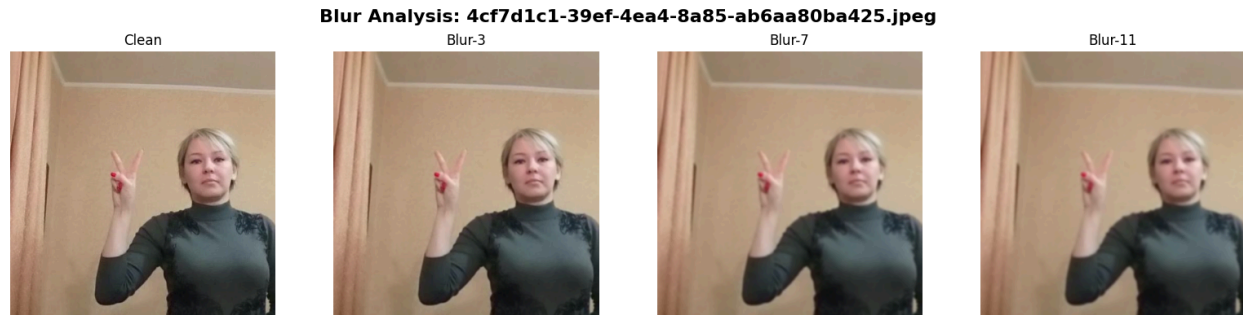
Examining per-class performance reveals interesting patterns. For ResNet-50 Frozen, classes such as 'mute' (F1=0.92) and 'dislike' (F1=0.82) performed well even without fine-tuning, while 'peace' (F1=0.28), 'palm' (F1=0.41), and 'three' (F1=0.36) performed poorly. This suggests that some gestures share visual features with common ImageNet objects, while others require domain-specific adaptation.

After fine-tuning, ResNet-50 achieved near-perfect per-class F1 scores across all 18 classes, with the lowest being 'four' at 0.95 and 'three' at 0.93. The SVM similarly achieved high per-class scores, with 'one' being the only class below 0.97 (F1=0.95). EfficientNet Fine-tune showed slightly lower per-class scores particularly for 'four' (F1=0.86) and 'one' (F1=0.86).

7.7 Class-wise Robustness to Gaussian Blur

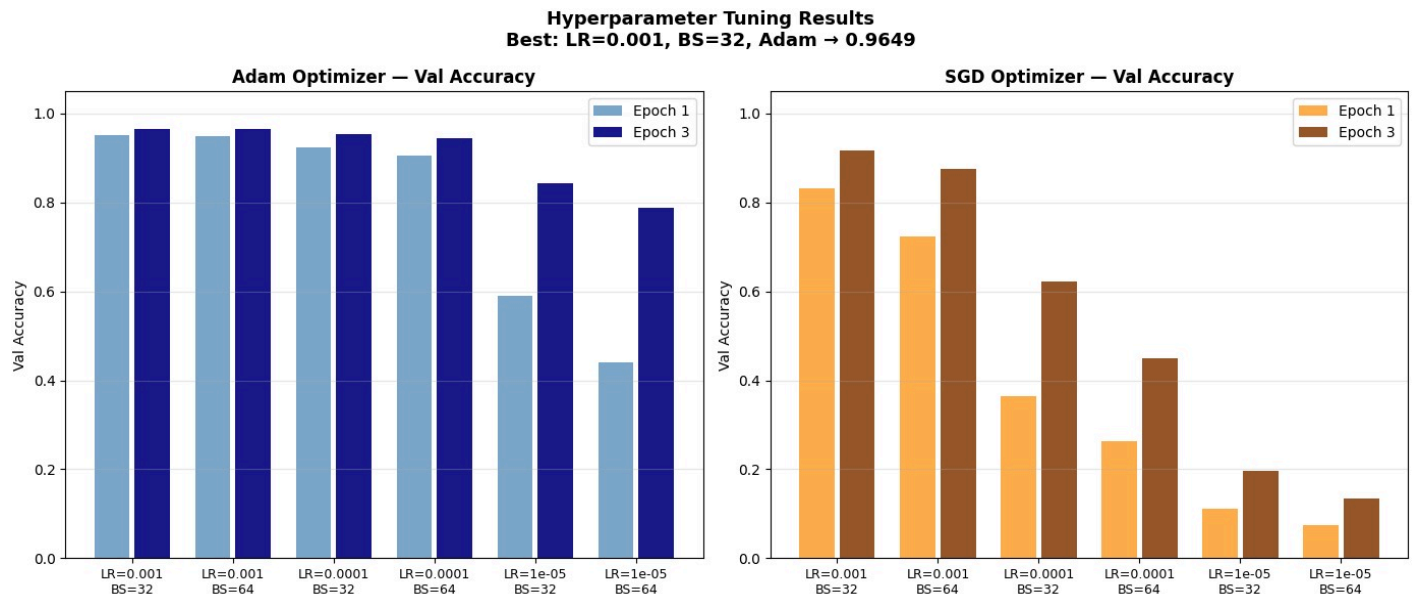
Class-wise robustness analysis was conducted by evaluating ResNet-50 Fine-tune on the test set under clean conditions and three levels of Gaussian blur (kernel sizes 3, 7, and 11), measuring accuracy separately for each of the 18 gesture classes. As shown in the heatmap, the model demonstrates strong robustness across nearly all classes, with most accuracies remaining above 0.94 even at the highest blur level. The "mute" class achieved perfect accuracy (1.00) across all blur levels, indicating that its distinctive closed-hand shape is highly resilient to visual degradation. On the other hand, the "peace" class showed the greatest sensitivity to blur, dropping from 0.93 under clean conditions to 0.89 at Blur-11, likely due to its reliance on fine-grained finger separation details that are easily obscured by blur. Overall, the results confirm that the fine-tuned model learns robust, blur-resistant representations for the majority of gesture classes, with only minor degradation observed even under severe corruption.





7.8 Hyperparameter Tuning Results

Hyperparameter tuning was performed by evaluating 12 combinations of learning rates (1e-3, 1e-4, 1e-5), batch sizes (32, 64), and optimizers (Adam, SGD with momentum=0.9) on the ResNet-50 Fine-tune architecture. Due to computational time constraints, each configuration was trained for 3 epochs and evaluated on the validation set. As illustrated in the bar charts, Adam consistently outperformed SGD across all learning rate and batch size combinations, converging significantly faster and reaching higher validation accuracy within the same number of epochs. Among all configurations, LR=0.001 with batch size 32 and the Adam optimizer achieved the highest validation accuracy of 0.9649 after 3 epochs, confirming it as the optimal configuration. SGD showed particularly poor performance at lower learning rates (1e-4 and 1e-5), struggling to converge meaningfully with 3 epochs. These results show that the final model training, where Adam with LR=1e-3 for the classification head and LR=1e-4 for the backbone, was selected as the primary optimizer configuration.



8. Conclusion

In this project, we compared two pretrained deep learning models, ResNet-50 and EfficientNet-B0, under frozen and fine-tuned training regimes, and evaluated an embedding-based SVM classifier for 18-class hand gesture recognition. The results clearly answer our research questions: fine-tuning significantly outperforms frozen backbone training for both architectures, with ResNet-50 Fine-tune achieving 97.16% test accuracy compared to 58.51% for its frozen counterpart, a gap that confirms the necessity of domain-specific adaptation. Among all approaches, the SVM classifier trained on PCA-reduced ResNet-50 embeddings achieved the highest clean accuracy of 97.67% and the strongest blur robustness, dropping only 1.11 percentage points at the highest corruption level, demonstrating that a classical machine learning classifier can match or surpass end-to-end deep learning when built on strong pretrained features. ResNet-50 consistently outperformed EfficientNet-B0 across all conditions, while EfficientNet-B0 proved particularly sensitive to blur in the frozen setting, dropping nearly 15 percentage points under heavy corruption. These findings highlight that fine-tuning is essential for transfer learning to be effective, that embedding-based approaches offer a competitive and robust alternative to end-to-end deep learning, and that architecture choice matters significantly under real-world image degradation.

For future work, extending robustness evaluation to other corruptions such as noise and occlusion, exploring deeper fine-tuning strategies by gradually unfreezing more layers, and evaluating these models on dynamic gesture recognition using video sequences would represent meaningful and practically relevant directions.

9. Workload Distribution

The workload for the project is distributed among the five group members according to the main tasks completed in the current stage.

Mehmet Dizdar took primary responsibility for the **ResNet-50** model. He worked on adapting the pretrained architecture to the hand gesture classification task, implementing the training pipeline, and conducting experiments for frozen-backbone and fine-tuning settings. He also contributed to comparing ResNet-50 results with the other approaches.

Aral Müftüoğlu took primary responsibility for the **dataset preparation and preprocessing pipeline**, including class-balanced sampling, train-validation-test splitting, image resizing, and augmentation setup. In addition, he contributed to the **ResNet-50 experiments**, particularly in data handling, preprocessing integration, and robustness analysis using blurred test images.

Yunus Emre Aslan took primary responsibility for the **embedding-based machine learning model**. He worked on extracting deep feature embeddings from the fine-tuned ResNet-50 model, applying PCA for dimensionality reduction, and training the SVM classifier. He also contributed to evaluating how the embedding-based approach compares with the end-to-end deep learning models.

Hüsnü Arkca took primary responsibility for the **EfficientNet-B0** model. He handled model setup, training configuration, hyperparameter testing, and evaluation under both clean and corrupted conditions. He also contributed to the comparison of EfficientNet-B0 against ResNet-50 and the embedding-based machine learning model.

Afşin Sıddık Atçeken took primary responsibility for the **evaluation and reporting stages** of the project, including metric calculation, confusion matrix analysis, and organization of experiment outputs. In addition, he contributed to the **EfficientNet-B0 experiments**, supporting training, validation-based comparison, and interpretation of the results.

All members: discussion of the project progress, interpretation of preliminary findings, and revision/editing of the final report.

This distribution allowed the team to work in parallel while maintaining consistency with the objectives of the project.

10. References

He, K., Zhang, X., Ren, S., & Sun, J. (2016). *Deep Residual Learning for Image Recognition*. Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR).

Tan, M., & Le, Q. V. (2019). EfficientNet: Rethinking model scaling for convolutional neural networks. Proceedings of the International Conference on Machine Learning (ICML).

HaGRID Classification 512p dataset, Kaggle.